

SAGONA ISLAND, NL, Canada

WMO: 714080

Lat: 47.367N

Lon: 55.795W

Elev: 196

StdP: 14.59

Time Zone: -3.50 (NAF)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.9	13.8	1.9	6.1	11.9	4.8	7.1	14.6	62.1	34.1	57.6	30.2	26.8	310	1.016

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	67.0	63.8	65.4	62.9	64.2	61.9	65.4	66.5	64.0	65.2	62.8	64.0	14.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.4	65.9	63.5	88.2	64.7	62.2	84.3	63.6	30.3	66.6	29.4	65.5	28.5	64.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
52.8	47.2	42.1	DB	7.0	72.3	4.4	3.7	3.8	75.0	1.2	77.1	-1.2	79.2	-4.4	81.9	
			WB	6.1	69.1	4.9	3.6	2.6	71.7	-0.3	73.8	-3.0	75.9	-6.5	78.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	27.0	26.2	29.4	35.6	42.5	49.3	57.3	61.2	56.1	48.0	40.3	32.3
	DBStd	12.77	6.32	6.74	5.77	4.17	3.90	3.96	3.56	2.85	4.16	4.85	5.62	6.31
	HDD50	3682	712	667	639	432	232	59	0	0	5	96	292	548
	HDD65	8325	1177	1087	1104	882	696	472	240	119	268	527	740	1013
	CDD50	837	0	0	0	0	1	38	226	349	187	34	2	0
	CDD65	3	0	0	0	0	0	0	1	2	0	0	0	0
	CDH74	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	20.6	26.7	25.1	22.8	20.2	17.0	15.2	14.5	15.8	18.9	22.0	23.8	25.9
(15) Precipitation	PrecAvg	65.4	5.9	5.6	5.2	4.9	4.8	4.2	4.8	4.1	6.1	6.9	6.9	6.3
	PrecMax	81.1	10.1	9.8	8.8	10.9	6.9	7.2	8.7	5.9	9.3	14.0	9.0	10.2
	PrecMin	57.3	3.5	3.0	2.9	2.0	1.3	1.7	2.2	2.8	3.1	3.7	3.4	4.3
	PrecStd	5.8	1.6	1.7	1.9	2.4	1.4	1.5	1.7	0.9	1.7	2.6	1.4	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	44.2	43.1	49.4	56.9	63.8	68.8	69.6	67.3	61.1	54.7	48.8
		MCWB	44.7	43.8	40.0	46.2	50.3	57.5	65.0	65.2	62.6	59.4	54.2	48.4
	2%	DB	40.8	40.5	40.1	45.8	52.7	59.2	65.4	67.6	64.8	58.5	52.5	45.7
		MCWB	40.4	39.8	37.6	43.5	48.3	56.0	62.7	64.3	62.4	57.3	52.1	45.1
	5%	DB	37.9	37.4	38.1	43.2	50.2	56.9	63.3	66.3	63.3	56.5	50.7	43.6
		MCWB	37.2	36.7	36.0	40.9	46.8	54.1	61.0	63.7	61.2	54.9	50.1	43.0
	10%	DB	35.5	34.7	36.2	41.1	48.1	54.8	61.8	65.1	61.6	54.7	48.4	40.9
		MCWB	34.7	33.7	34.4	39.1	45.2	52.6	60.0	62.8	59.5	52.8	47.4	40.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	43.9	41.2	47.4	52.4	60.2	67.5	67.5	65.1	61.1	54.8	49.9
		MCDB	45.7	44.3	42.1	48.2	54.9	62.1	70.1	68.7	65.8	61.5	55.0	50.0
	2%	WB	40.8	40.1	37.6	44.2	49.8	57.2	64.3	65.7	63.5	58.5	52.3	45.9
		MCDB	41.0	40.5	38.8	45.0	52.1	58.9	66.0	66.8	64.2	59.0	52.6	46.3
	5%	WB	37.6	36.1	35.7	41.8	48.0	55.0	62.4	64.7	61.8	56.0	50.5	43.5
		MCDB	37.9	36.5	36.9	43.0	50.3	56.4	63.6	65.9	62.9	56.7	50.9	44.0
	10%	WB	35.0	33.4	34.0	39.7	46.0	53.1	61.0	63.7	60.2	54.2	48.1	40.3
		MCDB	35.6	34.1	35.1	41.0	48.3	54.4	62.1	64.9	61.3	55.0	48.6	40.9
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.4	8.1	8.0	8.9	8.3	7.1	6.8	7.2	6.7	7.2	7.8
		MCDBR	12.1	11.3	9.5	11.0	12.7	11.8	9.5	8.1	8.3	7.8	9.5	10.3
		MCWBR	13.5	13.3	9.9	9.2	9.4	9.3	7.4	6.4	7.5	8.2	10.9	12.5
	5% WB	MCDBR	12.2	12.7	10.0	11.0	12.6	11.6	9.3	7.5	7.8	8.2	9.9	11.4
		MCWBR	13.5	13.3	10.1	9.7	9.8	9.5	7.8	6.4	7.4	8.9	11.1	12.7
(40) Clear-Sky Solar Irradiance	taub		0.317	0.314	0.330	0.370	0.375	0.399	0.424	0.404	0.369	0.345	0.320	0.308
	taud		2.345	2.436	2.435	2.347	2.397	2.360	2.336	2.389	2.475	2.494	2.491	2.402
	Ebn at Noon		239	271	283	280	282	275	266	266	267	256	240	230
	Edh at Noon		23	26	30	36	36	38	38	35	29	25	20	20
(44) All-Sky Solar Radiation	RadAvg		331	522	892	1325	1597	1630	1548	1381	1118	698	384	270
(45)	RadStd		24	58	63	118	124	167	120	84	69	51	46	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A

Nomenclature: See separate page